

Cultural Preservation Simulation

An Agentic Simulation of Cultural Preservation and Erosion

Track: B — Applied Agent Experience

Team: Iteoluwa Ibitoye

1. Architecture Diagram

The system follows a sequential three-agent pipeline:

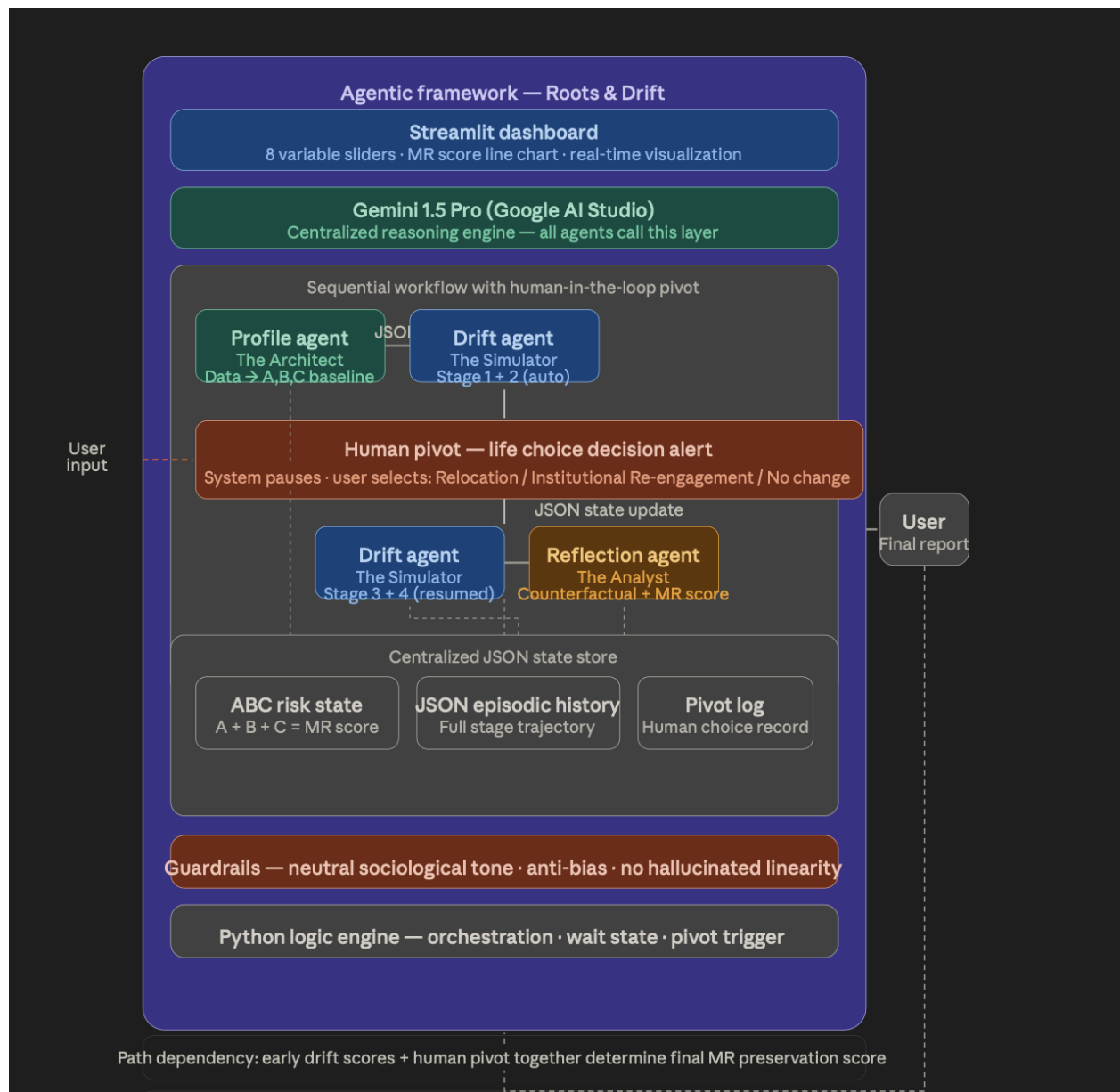
User Input → Profile Agent → Drift Simulation Agent → Reflection Agent → Final Report

The state is passed between agents as a JSON schema to ensure the system transfers structured, auditable data at each handoff. The user sets starting conditions by toggling the 8 variables before the simulation runs and receives the final report at the end. There's also a human-in-the loop intervention at age 18, where the user can make a qualitative choice (e.g., seeking a community anchor) that alters the subsequent drift trajectories.

The pipeline executes across six distinct phases:

1. **Geographic Research:** A background task that synthesizes diaspora data for the chosen destination (e.g., Quebec or New York City) to ground the simulation in reality.
2. **Baseline Profiling:** The **Profile Agent** (The Architect) generates starting scores based on user-defined inputs like age at migration and cultural anchor.
3. **Adolescent Drift:** The **Drift Agent** (The Simulator) models the cultural pressures of high school and peer socialization.
4. **The Pivot:** A manual intervention where the user selects a life path to mitigate or accept cultural drift.
5. **Adult Integration:** The Drift Agent resumes, factoring the user's pivot choice into the final stages of professional and social identity formation.
6. **Synthesis:** The **Reflection Agent** (The Analyst) produces a comprehensive report, comparing the actual outcome against counterfactual scenarios.

The LLM backbone is Groq's Llama 3.3 70B Versatile, with OpenAI GPT-4o Mini as a fallback. Both are called through the same OpenAI-compatible interface so the swap is invisible to the user.



2. Role Definitions

Profile Agent (The Architect)

This agent runs first. It takes the user's eight input variables alongside a geographic context summary that was generated silently before the session started, and figures out what a culturally grounded starting point looks like for this specific person in this specific city. The important thing it does that a simple lookup table cannot is calibrate scores based on how the variables interact with each other.

Attributes	Detail
Inputs	Age at migration, cultural anchor score, heritage language use, social homophily, neighborhood cultural density, institutional anchor presence, intergenerational transmission pressure, geographic context summary
Outputs	Starting scores (0-100) for four cultural dimensions, resilience anchors, risk factors, community density assessment, opening narrative paragraph
Key behavior	Calibrates scores based on interaction between variables, not independent averages. Geographic context directly influences social affiliation starting score.

Drift Simulation Agent (The Simulator)

This agent runs four times, once for each life stage. Each time it receives the current JSON state (which includes scores from the prior stage) and reasons about what changes given the pressures of that particular stage. The logic is the most important design feature here. The agent is instructed not to reset scores, so a person who loses language fluency in adolescence carries that loss into early adulthood. This models the real irreversibility of certain cultural losses.

After Stage 2, the simulation pauses and a human pivot choice gets written into the JSON state before Stage 3 runs. The agent receives this choice and has to factor it meaningfully into the trajectory. There is also a Cultural Re-awakening rule, if institutional anchors are present during Early Adulthood, scores can rise rather than fall, which reflects documented patterns of diaspora identity reconnection during later years, mostly college.

Attribute	Detail
Inputs	Current JSON state with prior stage scores, stage name and age range, pivot choice (for Stages 3 and 4), geographic context
Outputs	Updated scores for all four dimensions, primary driver identification, stage narrative, failure log entry if any dimension drops more than 15 points
Key behavior	Compounding logic enforced across stages. Volatility instruction prevents smooth linear drift. Re-awakening is possible in Stage 3 when institutional anchors are present.

Reflection Agent (The Analyst)

This agent runs once at the end. It gets the complete trajectory, all four stage histories, the failure log, the original profile, and the geographic context, and produces a synthesis report. The reason this is a separate agent rather than just a summary function is that it performs a qualitatively different kind of reasoning. It has to identify turning points, rank variables by influence, generate counterfactual scenarios, and connect the specific trajectory to broader sociological patterns. That is analysis, not summarization, and it benefits from being separated from the drift logic entirely.

Attribute	Detail
Inputs	Full JSON state, complete stage trajectory, original user configuration, geographic context, failure log
Outputs	Narrative summary, turning point identification, top three ranked drivers, two counterfactual analyses, preservation verdict, research note
Key behavior	Must rank the pivot against structural variables so users understand whether the pivot or the starting conditions drove the outcome. Counterfactuals must name specific alternative cities or pivot choices.

3. Coordination Logic

The coordination logic uses a Sequential Handoff with an Asynchronous Pause.

1. System Start + Profile Agent: The user initiates the session by configuring the 8 variables and all necessary information. The Profile Agent establishes the **InitialState** alongside the background Geographic Research
2. Automated Drift (Stages 1 & 2): The baseline json from the Profile Agent is passed to the Drift Agent, which simulates Arrival and Adolescence based purely on environmental variables.
3. The "Life Event" Pivot (Human Intervention):
 - The system enters a "Wait State" after Stage 2.
 - The user is shown their current trajectory and must select one "Life Choice" (e.g., Proactive Cultural Traveler/ Social Relocation, Cultural Orgs,...) to modify the pressure for the remaining stages.
4. Resumption & Cumulative Scoring: The Drift Agent receives the updated variable and processes the final two stages, ensuring the new scores build upon the existing "drifted" state rather than starting fresh.
5. Termination: The process stops once the Reflection Agent generates the final report and Preservation Score.

Step	Who Acts	What Happens	Human Involved?
1	System	Geographic research runs silently in background	No
2	Profile Agent	Builds cultural baseline from 8 variables + geographic context	No
3	Drift Agent	Simulates Stage 1 (Arrival) and Stage 2 (Adolescence)	No
4	System + User	Simulation pauses. User selects a life pivot from four options.	Yes, required
5	Drift Agent	Simulates Stage 3 (Early Adulthood) and Stage 4 (Established Life) with pivot factored in	No
6	Reflection Agent	Generates final tabbed report with narrative, drivers, counterfactuals	No
7	User	Reviews report, exports JSON, optionally restarts with new variables	Yes — optional

4. Tools, Memory, and Data Design

4.1 Technology Stack

The components were selected to provide low-latency inference and rapid development without compromising security.

Component	Tool / Library	Reasoning
LLM (Primary)	Groq Llama 3.3 70B	Offers speedy inference and JSON mode support essential for high-speed simulation.
LLM (Fallback)	OpenAI GPT-4o Mini	Provides a reliable secondary model via an OpenAI-compatible interface with no code changes.
Frontend	Streamlit	Handles session state management natively, making it ideal for multi-stage sequential flows.
Visualization	Plotly	Generates interactive line charts that update in real-time as the identity trajectory evolves.

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LLM (Primary)	Groq Llama 3.3 70B	Offers speedy inference and JSON mode support essential for high-speed simulation.

4.2 Memory and State Design

The simulation does not rely on a global chat history, which could lead to model drift or hallucinations. Instead, it uses **Explicit State-Passing**.

- **State Persistence:** The entire **SimulationHistory** object is stored in the Streamlit `session_state`. Every agent call receives the full JSON object, ensuring that the "Profile" created in Chapter 1 is exactly what the "Reflection Agent" analyzes in the final report.
- **Auditability:** Because the state is a single, structured JSON object, the user can inspect the "State Store" at any point to see precisely how and why a specific agent modified a score.

The JSON Schema includes:

- **scores:** Current dimension values updated after each lifecycle stage.
- **resilience_anchors/risk_factors:** Strategic tags identified by the Profile Agent to guide the Drift Agent's logic.
- **stage_history:** A full record of every agent's narrative and score changes.
- **failure_log:** Tracking "Tipping Points" where a dimension drops by **15+ points** in a single stage.

4.3 MR Score Calculation & Thresholds

To quantify the "Magnitude of Cultural Retention" (MR), the system uses a weighted composite score that treats internal values and external behaviors differently.

The Formula: $MR = (Language + Cultural Practice + ((Social Affiliation + Self-Presentation) / 2)) / 3$

- **Primary Weighting:** Language and Cultural Practice are treated as independent, primary pillars of identity.
- **Social Weighting:** Social Affiliation and Self-Presentation are averaged into a single "Social Identity" score before being weighted against the primary pillars.

Calculated Thresholds:

- **MR > 65: Strong Preservation** — High cultural resilience.

- **MR 40–65: Moderate Erosion** — Hybrid or hyphenated identity state.
- **MR < 40: Critical Erosion** — Deep assimilation into the host culture.

4.4 Data Lifecycle

The flow of data ensures that sociological variables are consistently applied throughout the character's life.

1. **User Input:** Raw variables (age, anchor, density) are captured.
2. **Profile Output:** The "Architect" establishes **baseline_scores**.
3. **Drift Output:** The "Simulator" appends to the **trajectory_array** across four life stages.
4. **Reflection Output:** The "Analyst" generates the **final_report** and calculates the aggregate MR.

5. Prototype/Partial Build

The prototype is a fully functional Streamlit application with structured JSON data flowing between agents to ensure a cohesive narrative. The user journey begins with a setup screen where eight sociological variables are configured. Behind the scenes, the Geo-Researcher performs initial context gathering before the Profile Agent builds the baseline identity, displaying starting scores and a tailored "Arrival" narrative. The Drift Agent then automates the first two life stages, with an interactive Plotly chart updating in real-time to visualize identity shifts.

A critical architectural feature is the mandatory pause at age 18. Here, the simulation waits for a human "Pivot Choice," which is then injected into the state store to guide the final two drift stages. The process concludes with a tabbed Reflection Report that provides deep-dive analysis into turning points and counterfactual scenarios. Early testing confirmed that the system is sensitive to these inputs; for example, choosing a "Community Anchor" versus "Professional Assimilation" produced meaningfully divergent scores in later stages, proving the Drift Agent is reasoning about the pivot rather than following a pre-set script.

Deploy

Configure the Simulation

Define the migration journey. The more specific the location, the more grounded the story.

The journey

Origin

Country of origin
e.g. Nigeria

City or region of origin
e.g. Lagos

Heritage language spoken at home
e.g. Yoruba, Igbo, Spanish

Destination

Destination country
e.g. United States

Destination city / state
e.g. Columbus, Ohio

Sociological variables

Age at migration
7

Neighborhood cultural density
15

Cultural anchor (depth of immersion in origin culture)
5

Heritage language use frequency at home
4

Social homophily (in-ethnic peer preference)
5

Institutional anchor present (cultural church, association, etc.)
☒

Intergenerational transmission pressure (parental enforcement)
4

Please fill in all location and language fields.

Roots & Drift

A cultural preservation simulation

AI providers

Groq (primary)

OpenAI (fallback)

Journey

Shanghai, China

Pittsburgh, United States

Mandarin

Restart

Geographic Research

Researching the Community Context

Analyzing the China diaspora community in Pittsburgh...

Community context: Shanghai → Pittsburgh

Via Groq

The China diaspora in Pittsburgh is estimated to be around 10,000-15,000 individuals, with a notable concentration in the Squirrel Hill neighborhood, which is home to a variety of Asian restaurants and cultural shops. The Pittsburgh Chinese Church and the Chinese Association for Science and Technology serve as cultural institutions for the community, while the University of Pittsburgh's Confucius Institute also plays a significant role in promoting Chinese language and culture. The immigrant experience for this group in Pittsburgh is generally marked by a mix of warmth and challenges, with many finding employment opportunities in the city's universities and medical centers, but also facing cultural and linguistic barriers. While there are support systems in place, such as the Pittsburgh Chinese Community Center, the community may still feel somewhat isolated due to the relatively small size of the diaspora compared to other US cities. The Squirrel Hill neighborhood, in particular, has emerged as a cultural hub for the community, with its annual Lunar New Year parade and festival drawing large crowds. Overall, while my knowledge of the specific nuances of the Pittsburgh Chinese diaspora is limited, it appears that the community has established a vibrant presence in the city, with a range of cultural institutions and events to support its members.

Continue to Profile Agent →

Roots & Drift

A cultural preservation simulation

AI providers

Groq (primary)

OpenAI (fallback)

Live MR Score

80.8 / 100

Strong preservation

Language: 80

Cultural Practice: 85

Social Affiliation: 75

Self Preservation: 80

Journey

Shanghai, China

Pittsburgh, United States

Mandarin

Restart

Chapter 7 — Reflection Agent

The Final Report

The Analyst synthesizes the full trajectory into research-grade insight.

80.8 / 100

Final Cultural Preservation Score — Strong preservation

Score (0-100)

Baseline

Arrival

Adolescence

Life stage

Early Adulthood

Established Life

Language

Cultural practice

Social affiliation

Self preservation

MR composite

Narrative

Turning Points & Drivers

Counterfactuals

Research & Data

Narrative summary

Via Groq

As you navigated your life in Pittsburgh, you experienced a mix of challenges and opportunities that shaped your cultural identity. Your strong family ties to Mandarin and Chinese culture helped you maintain your heritage, but you faced cultural and linguistic barriers in daily life. Joining a co-ethnic cultural organization in Early Adulthood was a pivotal decision that facilitated a Cultural Re-awakening, allowing you to engage with heritage peers and participate in cultural events. This resurgence in cultural practice and language skills helped you establish a strong sense of belonging in Pittsburgh. Your involvement with the local Chinese community has become a cornerstone of your identity, with the city's vibrant Squirrel Hill neighborhood serving as a cultural hub.

Preservation verdict

Your cultural identity has experienced Strong Preservation, as you have maintained a strong connection to your heritage language and culture throughout your life in Pittsburgh, thanks to a combination of strong family ties, institutional anchors, and community involvement.

Community context used

The China diaspora in Pittsburgh is estimated to be around 10,000-15,000 individuals, with a notable concentration in the Squirrel Hill neighborhood, which is home to a variety of Asian restaurants and cultural shops. The Pittsburgh Chinese Church and the Chinese Association for Science and Technology serve as cultural institutions for the community, while the University of Pittsburgh's Confucius Institute also plays a significant role in promoting Chinese language and culture. The immigrant experience for this group in Pittsburgh is generally marked by a mix of warmth and challenges, with many finding employment opportunities in the city's universities and medical centers, but also facing cultural and linguistic barriers. While there are support systems in place, such as the Pittsburgh Chinese Community Center, the community may still feel somewhat isolated due to the relatively small size of the diaspora compared to other US cities. The Squirrel Hill neighborhood, in particular, has emerged as a cultural hub for the community, with its annual Lunar New Year parade and festival drawing large crowds. Overall, while my knowledge of the specific nuances of the Pittsburgh Chinese diaspora is limited, it appears that the community has established a vibrant presence in the city, with a range of cultural institutions and events to support its members.

Run a new simulation

Justification for Multi-Agent System

1. Human intervention cannot exist smoothly in a single call: The user makes a choice that changes what happens next but a single prompt has to know the ending before it starts. The multi-agent architecture with a wait state is a great way to implement a genuine human-in-the-loop mechanic.
2. Each agent fails independently and can be debugged: If the single-call version produces a bad output, you have no idea which part of the reasoning went wrong but in the pipeline, if the Drift Agent produces implausible scores you can inspect the exact input it received and the exact output it returned.
3. Dividing tasks produces better results: The Drift Agent is modeling pressure and change while The Reflection Agent is interpreting a completed trajectory and generating counterfactuals. These are different cognitive tasks so asking one model to do both simultaneously in one call produces worse output than letting a dedicated agent see the full completed trajectory before it starts analyzing.

6. Evaluation Tests

The evaluation is centered on five distinct test scenarios designed to test specific hypotheses about the simulation's reasoning capabilities. I would be looking for sociological plausibility and evidence that the agents are meaningfully responding to variable changes rather than producing generic drift patterns.

Case ID	Scenario	Input Profile Highlights	Hypothesis	Success Criteria	Measure
TC-01	The Late Arrival	Age: 16. Anchor: 8/10. Density: Low.	High early-life anchors at late arrival will protect language fluency despite low neighborhood support.	Language score at age 25 $\geq$ 60.	Stage 4 score check

TC-02	The Child Migrant	Age: 9. Anchor: 3/10. Density: Low.	Early arrival with weak cultural anchors produces near-complete language erosion by adulthood.	Language score at age 25 < 25.	Stage 4 score check
TC-03	Institutional Buffer	Age: 12. Run twice: Institutional Anchor Present vs. Absent .	An institutional anchor (e.g., a cultural center) protects practice even in low-density neighborhoods.	Cultural Practice score is ≥ 15 points higher in the "Present" run.	Delta comparison across 2 runs
TC-04	Pivot Divergence	Same baseline. Pivot A: Community. Pivot B: Career.	The human pivot choice produces divergent life trajectories, not just cosmetic narrative changes.	At least two dimensions diverge by ≥ 12 points by Stage 4.	Multi-dimension delta check
TC-05	Structural Inertia	Age: 9. Density: Very Low. All protective variables at min.	When structural conditions strongly favor assimilation, even a protective pivot cannot reverse the drift.	Final MR remains < 45; Pivot impact on any dimension is < 8 points.	MR score + per-dimension delta

These numbers were created to give the **Reflection Agent** a ruler/baseline to measure the user's life. TC-03 and TC-04 are very important because they require running the simulation twice with a controlled variable change, which is the only way to isolate the effect of a single factor. If the two runs produce nearly identical outputs, that is evidence the agent is not actually reasoning about the variable but it is just producing consistent-sounding text. TC-05 is the hardest case because it tests whether the system can produce a pessimistic realistic outcome rather than always trending toward recovery. A system that always ends with moderate preservation regardless of inputs is not a simulation, it is just a narrative generator.

The Reflection Agent will be evaluated separately on counterfactual accuracy: for TC-03, the agent must identify neighborhood density as the primary driver and name a specific city with higher density as the alternative in Counterfactual A. For TC-04, it must name the unchosen pivot as Counterfactual B and estimate the score difference. If the counterfactuals are vague or generic they fail regardless of the score outcomes.

7. Risk and Governance Plan

Risk 1: Hallucinated linearity

LLMs naturally produce smooth, coherent stories, which means the Drift Agent may default to dropping scores by a consistent amount each stage rather than modeling the non-linear reality of cultural assimilation — where some stages are volatile and others are stable. The mitigation is the Volatility Instruction in the Drift Agent prompt, which explicitly asks for sudden drops, stagnation periods, and possible recovery events. TC-05 is specifically designed to catch this: if a profile with maximum structural pressure shows a smooth gradual decline rather than a sharp early drop followed by stabilization, the agent has hallucinated linearity.

Risk 2: Cultural overgeneralization

The model may apply Nigerian or broadly West African cultural patterns to profiles from other origin countries, particularly for less-represented diaspora communities. This is a real limitation of what the LLM knows. The Drift Agent already includes a failure log that should flag low-confidence cultural reasoning, but this may not catch subtle errors. For example, a Mexican immigrant and a Nigerian immigrant both in Houston might receive outputs that match a large diaspora community without accounting for the very different cultural institutions, linguistic preservation rates, and social reception patterns that characterize each group. This is why the geographic research was included, to add more external context to each simulation but a mitigation for Phase 3 is to test several varying cultural profiles in the evaluation runs and compare the geographic context paragraphs for specificity. If Houston's Mexican community and Houston's Nigerian community produce identical-sounding community context summaries, that is a failure mode to document in the failure log.

Risk 3: Data privacy

Users may enter information about themselves or family members that is sensitive like immigration history, family dynamics, cultural identity. The simulation stores all of this in Streamlit session state, which is cleared when the browser session ends so nothing is logged, persisted to a database, or sent anywhere other than the LLM API call. The API calls do include the user's inputs in the prompt, so the data is processed by Groq or OpenAI under their standard API terms. This would be disclosed to users before they run the simulation

Risk 4: Evaluation Overfitting

The evaluation plan's current success criteria are thresholds all chosen by me, the developer. The LLM could learn to hit these numeric targets without sociological plausibility e.g., moving scores just enough to pass but with nonsensical narratives. To govern this, the Reflection Agent is evaluated separately on the specificity of its counterfactuals. Vague or generic responses result in an automatic failure of the test case, regardless of the numeric score.

8. Contribution Update

As an individual submission, all roles are led by Iteoluwa Ibitoye.

- **Phase 2 Completion:** Orchestration logic and JSON schemas finalized. Interaction map finalized for Track B.
- **Phase 3 Schedule:** Conclude system build and execute the 5 evaluation scenarios and generate reports.

9. AI Appendix- Claude

Claude (Anthropic, claude.ai) was used throughout Phase 2 as a drafting tool mainly for the Streamlit prototype code. All prompt language, variable choices, and architectural decisions were reviewed and revised by me.

Interaction 1: Architectural direction and agent design

Prompt given: A description of the simulation concept and confirmation about whether a multi-agent system was actually necessary or whether a single LLM call could do the same thing.

What Claude produced: An explanation of four justifications for multi-agent design,

What was accepted: The four-part justification framework was accepted and became the basis for the architecture rationale in the Phase 2 report.

Interaction 2: Building the Streamlit prototype

Prompt given: The existing early-stage Streamlit code was shared along with a request to fix a broken `call_gemini` function that was returning 404 errors, and to rebuild the app using a more reliable LLM provider.

What Claude produced: A diagnosis of the root cause (the `google-generativeai` package being fully deprecated), a recommendation to switch to Groq with OpenAI as fallback, and a full rewrite of the app using the `openai`-compatible interface for both providers.

What was accepted: The Groq-first, OpenAI-fallback architecture was accepted. The `call_llm` function design with `json_mode` as a parameter was accepted. The `.env` file approach for API key management was accepted and implemented.

What was revised: The author made the decision to remove the geographic research stage from the visible UI so it runs silently in the background. Claude had initially built it as a separate screen the user had to click through. The author felt it interrupted the story flow and made it feel like an administrative step rather than seamless context-setting. The author also adjusted the pivot card layout and the sidebar live score display based on personal preference for how the interface should feel.

Interaction 3: Geographic grounding feature

Prompt given: A question about whether adding origin city, destination city, and heritage language would make the simulation feel more story-like and grounded, and steps to take to add a live web search to research real diaspora communities.

What Claude produced: Two options — live Gemini web search grounding, or Groq/OpenAI training knowledge used with a specific geographic prompt. After live search was chosen and then repeatedly failed due to API version issues, Claude recommended switching to training knowledge with a well-structured geographic research prompt.

What was accepted: The geographic research prompt structure was accepted. The fallback from live search to training knowledge was accepted as the right practical decision given the API constraints.